Worksheet 1.4 — Solving by Elimination

Grade 10 Academic (MPM2D) · Unit 1: Linear Systems

The Big Ideas

Elimination adds or subtracts the equations so that one variable cancels.

- Line the equations up as $Ax + By = C$.
- If a variable's coefficients are *opposites*, **add**; if they are *equal*, **subtract**.
- If neither matches, **multiply** one (or both) equations to create a matching pair first.
- Solve for the remaining variable, then back-substitute.

Concept Spotlight

The idea

Elimination adds or subtracts the two equations to cancel one variable. If needed, multiply an equation first so a pair of coefficients line up.

The steps

Stack the equations, scale one or both so a variable's coefficients are equal or opposite, add or subtract to eliminate it, solve, then back-substitute.

Choosing a method

Elimination is often fastest when no variable is already isolated, especially when the coefficients share a simple multiple.

Worked Examples

Example 1 — Add to eliminate

Solve:

$$\begin{aligned} x+y &= 10 \quad ① \\ x-y &= 4 \quad ② \end{aligned}$$

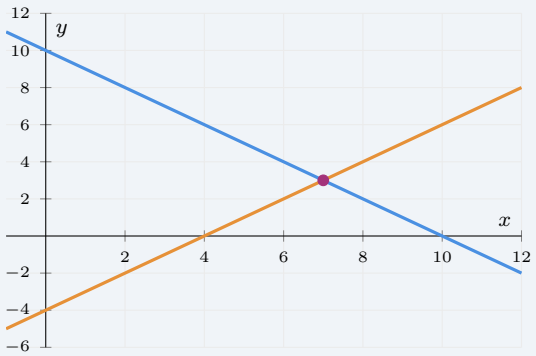
Solution. **Step 1:** Add the equations to eliminate y:

$$\begin{aligned} & x+\cancel{y}=10 \\ +(\ & x-\cancel{y}=4\ ) \\ \hline & (x+x)+(y-y)=10+4 \\ & 2x=14 \\ & \tfrac{2x}{2}=\tfrac{14}{2} \\ & x=7 \end{aligned}$$

Step 2: Substitute $x=7$ into ①:

$$\begin{aligned} 7+y &= 10 \\ y &= 10-7 \\ y &= 3 \end{aligned}$$

Conclusion: $(x,y)=(7,3)$. Graphing $y=10-x$ and $y=x-4$ shows the lines crossing at $(7,3)$.



Example 2 — Add (opposite y)

Solve:

$$\begin{aligned} 2x + y &= 7 \quad ① \\ x - y &= 2 \quad ② \end{aligned}$$

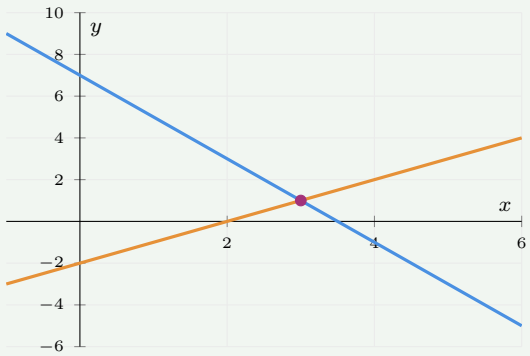
Solution. **Step 1:** Add the equations to eliminate y:

$$\begin{aligned} & 2x + \cancel{y} = 7 \\ +(\; & x - \cancel{y} = 2 \,) \\ \hline & (2x + x) + (y - y) = 7 + 2 \\ & 3x = 9 \\ & \tfrac{3x}{3} = \tfrac{9}{3} \\ & x = 3 \end{aligned}$$

Step 2: Substitute $x = 3$ into ②:

$$\begin{aligned} 3 - y &= 2 \\ -y &= 2 - 3 \\ -y &= -1 \\ \tfrac{-y}{-1} &= \tfrac{-1}{-1} \\ y &= 1 \end{aligned}$$

Conclusion: $(x, y) = (3, 1)$. Graphing $y = 7 - 2x$ and $y = x - 2$ confirms the crossing point.



Example 3 — Opposite coefficients

Solve:

$$\begin{aligned} 3x + 2y &= 12 \quad ① \\ 3x - 2y &= 0 \quad ② \end{aligned}$$

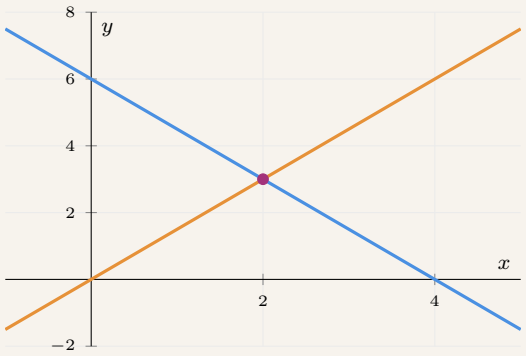
Solution. **Step 1:** Add the equations to eliminate y (the y-coefficients are opposites):

$$\begin{aligned} & 3x + \cancel{2y} = 12 \\ +\,(& 3x - \cancel{2y} = 0\,) \\ \hline & (3x + 3x) + (2y - 2y) = 12 + 0 \\ & 6x = 12 \\ & \tfrac{6x}{6} = \tfrac{12}{6} \\ & x = 2 \end{aligned}$$

Step 2: Substitute $x = 2$ into ①:

$$\begin{aligned} 3(2) + 2y &= 12 \\ 6 + 2y &= 12 \\ 2y &= 12 - 6 \\ 2y &= 6 \\ \tfrac{2y}{2} &= \tfrac{6}{2} \\ y &= 3 \end{aligned}$$

Conclusion: $(x, y) = (2, 3)$. Graphing $y = 6 - 1.5x$ and $y = 1.5x$ shows the lines meeting at $(2, 3)$.



Example 4 — Multiply one equation

Solve:

$$\begin{aligned} 2x + y &= 5 \quad ① \\ x + 3y &= 10 \quad ② \end{aligned}$$

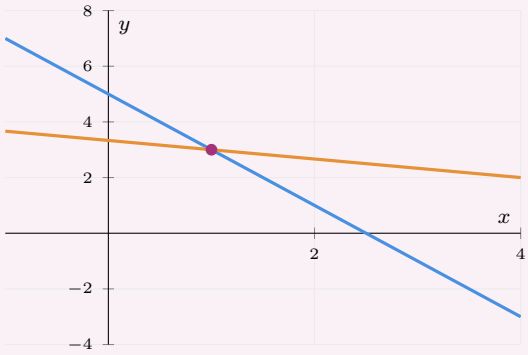
Solution. **Step 1:** Multiply ① by 3, then subtract ②:

$$\begin{aligned} & 6x + \cancel{3y} = 15 \\ -(\ & x + \cancel{3y} = 10\ ) \\ \hline & (6x - x) + (3y - 3y) = 15 - 10 \\ & 5x = 5 \\ & \tfrac{5x}{5} = \tfrac{5}{5} \\ & x = 1 \end{aligned}$$

Step 2: Substitute $x = 1$ into ①:

$$\begin{aligned} 2(1) + y &= 5 \\ 2 + y &= 5 \\ y &= 5 - 2 \\ y &= 3 \end{aligned}$$

Conclusion: $(x, y) = (1, 3)$. Graphing $y = 5 - 2x$ and $y = \frac{10-x}{3}$ confirms the crossing point.



Example 5 — Multiply both equations

Solve:

$$\begin{aligned} 2x + 3y &= 7 \quad ① \\ 3x + 2y &= 8 \quad ② \end{aligned}$$

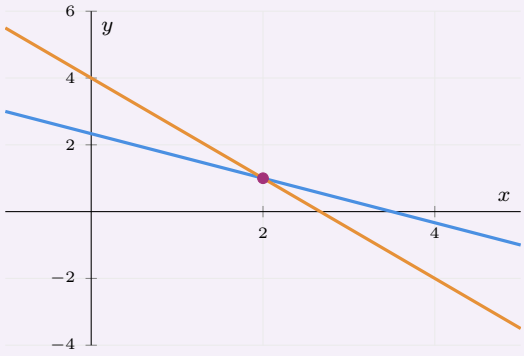
Solution. **Step 1:** To eliminate x, multiply ① by 3 and ② by 2, then subtract:

$$\begin{aligned} & \cancel{6x} + 9y = 21 \\ -(& \cancel{6x} + 4y = 16) \\ \hline & (9y - 4y) = (21 - 16) \\ & 5y = 5 \\ & \tfrac{5y}{5} = \tfrac{5}{5} \\ & y = 1 \end{aligned}$$

Step 2: Substitute $y = 1$ into ①:

$$\begin{aligned} & 2x + 3(1) = 7 \\ & 2x + 3 = 7 \\ & 2x = 7 - 3 \\ & 2x = 4 \\ & \tfrac{2x}{2} = \tfrac{4}{2} \\ & x = 2 \end{aligned}$$

Conclusion: $(x, y) = (2, 1)$. Graphing $y = \frac{7-2x}{3}$ and $y = \frac{8-3x}{2}$ shows the lines crossing at $(2, 1)$.



Example 6 — Infinitely many

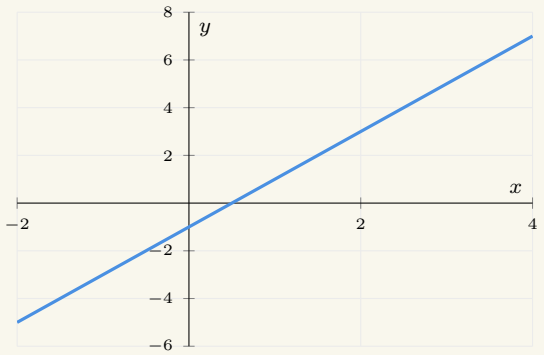
Solve:

$$\begin{aligned} 4x - 2y &= 2 \quad ① \\ 2x - y &= 1 \quad ② \end{aligned}$$

Solution. **Step 1:** Multiply ② by 2, then subtract it from ①:

$$\begin{array}{r} 4x - 2y = 2 \\ -\,(\,4x - 2y = 2\,) \\ \hline 0 = 2 - 2 \\ 0 = 0 \end{array}$$

Conclusion: $0 = 0$ is always true, so there are infinitely many solutions (both equations describe the same line). Graphing $4x - 2y = 2$ as $y = 2x - 1$ shows both equations are the exact same line.



Example 7 — No solution

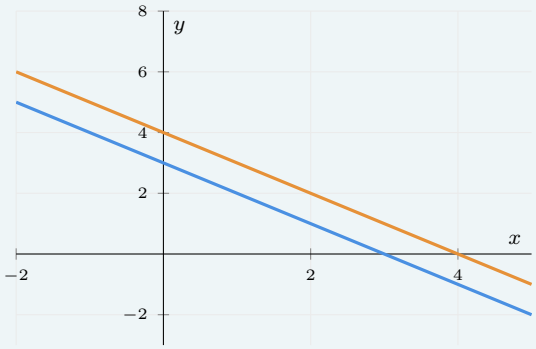
Solve:

$$\begin{aligned} x + y &= 3 \quad ① \\ 2x + 2y &= 8 \quad ② \end{aligned}$$

Solution. **Step 1:** Multiply ① by 2, then subtract ②:

$$\begin{aligned} \cancel{2x} + \cancel{2y} &= 6 \\ -\,(\,\cancel{2x} + \cancel{2y} &= 8\,) \\ \hline 0 &= 6 - 8 \\ 0 &= -2 \end{aligned}$$

Conclusion: $0 = -2$ is false, so there is no solution (the lines are parallel). Graphing $y = 3 - x$ and $y = 4 - x$ shows why: both have slope -1 but different y-intercepts, so the lines are parallel and never cross.



Example 8 — Word problem

Two adults and three children pay \$31; one adult and two children pay \$18. Find each price.

Solution. Let a be the adult price and c the child price, so the system is

$$\begin{aligned} 2a + 3c &= 31 \quad ① \\ a + 2c &= 18 \quad ② \end{aligned}$$

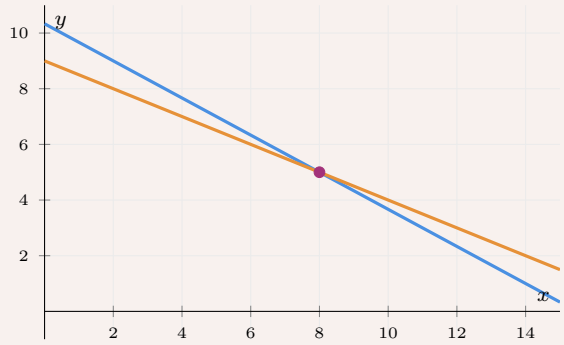
Step 1: Multiply ② by 2, then subtract ①:

$$\begin{aligned} & \cancel{2a} + 4c = 36 \\ -\,(& \cancel{2a} + 3c = 31 \,) \\ \hline & (4c - 3c) = (36 - 31) \\ & c = 5 \end{aligned}$$

Step 2: Substitute $c = 5$ into ②:

$$\begin{aligned} a + 2(5) &= 18 \\ a + 10 &= 18 \\ a &= 18 - 10 \\ a &= 8 \end{aligned}$$

Conclusion: $(a, c) = (8, 5)$: an adult ticket costs \$8 and a child ticket costs \$5. Graphing $c = \frac{31-2a}{3}$ and $c = \frac{18-a}{2}$ (adult price a on the horizontal axis, child price c on the vertical) shows the lines crossing at $(8, 5)$.



Example 9 — Multiply both (harder)

Solve:

$$\begin{aligned} 3x + 2y &= 4 \quad ① \\ 2x + 5y &= -1 \quad ② \end{aligned}$$

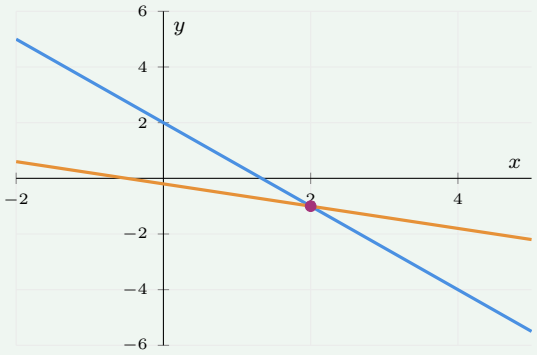
Solution. **Step 1:** To eliminate y, multiply ① by 5 and ② by 2, then subtract:

$$\begin{aligned} & 15x + \cancel{10y} = 20 \\ -\,(& 4x + \cancel{10y} = -2\,) \\ \hline & (15x - 4x) = (20 - (-2)) \\ & 11x = 22 \\ & \tfrac{11x}{11} = \tfrac{22}{11} \\ & x = 2 \end{aligned}$$

Step 2: Substitute $x = 2$ into ①:

$$\begin{aligned} 3(2) + 2y &= 4 \\ 6 + 2y &= 4 \\ 2y &= 4 - 6 \\ 2y &= -2 \\ \tfrac{2y}{2} &= \tfrac{-2}{2} \\ y &= -1 \end{aligned}$$

Conclusion: $(x, y) = (2, -1)$. Graphing $y = \frac{4-3x}{2}$ and $y = \frac{-1-2x}{5}$ confirms the crossing point.



Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Solve by elimination: $x + y = 8$ and $x - y = 2$.

Question 2

Solve by elimination: $2x + y = 9$ and $x - y = 0$.

Question 3

Solve by elimination: $x + 2y = 7$ and $x - 2y = 1$.

Question 4

Solve by elimination: $3x + y = 10$ and $x + y = 4$.

Question 5

Solve by elimination: $2x + 3y = 12$ and $2x - y = 4$.

Question 6

Solve by elimination: $x + 3y = 9$ and $2x + y = 8$.

Question 7

Solve by elimination: $3x + 2y = 7$ and $5x + 2y = 11$.

Question 8

Solve by elimination: $2x + 5y = 1$ and $3x - 5y = 14$.

Question 9

Solve: $2x - y = 3$ and $4x - 2y = 6$. How many solutions?

Question 10

Solve: $x + y = 5$ and $2x + 2y = 12$. How many solutions?

Question 11

Three pens and two erasers cost \$13; one pen and two erasers cost \$7. Find each price.

Question 12

Solve by elimination: $4x + 3y = 10$ and $2x + y = 4$.

Question 13 — Challenge

Solve by elimination: $3x + 4y = 5$ and $2x + 3y = 4$.

Question Recap & Answer Key

Question Recap

1. Solve by elimination: $x + y = 8$ and $x - y = 2$.
2. Solve by elimination: $2x + y = 9$ and $x - y = 0$.
3. Solve by elimination: $x + 2y = 7$ and $x - 2y = 1$.
4. Solve by elimination: $3x + y = 10$ and $x + y = 4$.
5. Solve by elimination: $2x + 3y = 12$ and $2x - y = 4$.
6. Solve by elimination: $x + 3y = 9$ and $2x + y = 8$.
7. Solve by elimination: $3x + 2y = 7$ and $5x + 2y = 11$.
8. Solve by elimination: $2x + 5y = 1$ and $3x - 5y = 14$.
9. Solve: $2x - y = 3$ and $4x - 2y = 6$. How many solutions?
10. Solve: $x + y = 5$ and $2x + 2y = 12$. How many solutions?
11. Three pens and two erasers cost \$13; one pen and two erasers cost \$7. Find each price.
12. Solve by elimination: $4x + 3y = 10$ and $2x + y = 4$.
13. Solve by elimination: $3x + 4y = 5$ and $2x + 3y = 4$.

✓ Answer Key

1. Add: $2x = 10 \Rightarrow x = 5$, $y = 3$: $(5, 3)$.
2. Add: $3x = 9 \Rightarrow x = 3$, $y = 3$: $(3, 3)$.
3. Add: $2x = 8 \Rightarrow x = 4$, $y = 1.5$: $(4, 1.5)$.
4. Subtract: $2x = 6 \Rightarrow x = 3$, $y = 1$: $(3, 1)$.
5. Subtract: $4y = 8 \Rightarrow y = 2$, $x = 3$: $(3, 2)$.
6. $\times 2$ first, subtract: $5y = 10 \Rightarrow y = 2$, $x = 3$: $(3, 2)$.
7. Subtract: $2x = 4 \Rightarrow x = 2$, $y = 0.5$: $(2, 0.5)$.
8. Add: $5x = 15 \Rightarrow x = 3$, $y = -1$: $(3, -1)$.
9. Infinitely many (second $= 2\times$ first).
10. No solution ($2x + 2y$ can't equal both 10 and 12).
11. Subtract: 2 pens $= 6 \Rightarrow$ pen \$3, eraser \$2.
12. $\times 2$ second, subtract: $y = 2$, $x = 1$: $(1, 2)$.
13. $\times 3, \times 4$: $9x + 12y = 15$, $8x + 12y = 16$; subtract $x = -1$, $y = 2$: $(-1, 2)$.